

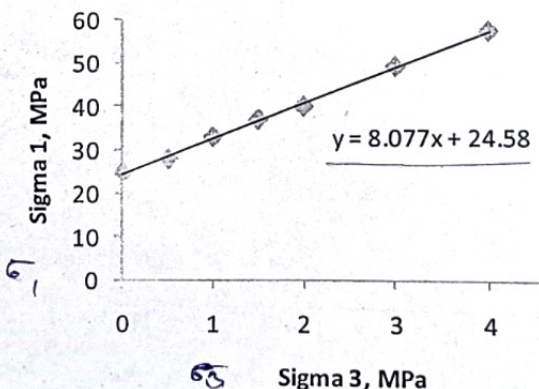
ROCK MECHANICS PRACTICAL EXAMINATION (MNC 209)

IV- Sem. B. Tech (ME) 2022-23

Time: 60 min

Instructions: Answer ALL questions

Total Marks: 30

Q. No.	Questions	Marks																
1.	A cylindrical specimen of sand stone of 50mm diameter and 125mm length, tested in uniaxial compression gave the following results: Load of failure = 30 tonne Longitudinal deformation upto = 1.2mm Assuming load deformation curve to be straight line upto failure, find the modulus of elasticity of the sand stone.	5																
2.	During Brazillian test a rock specimen of diameter 5 cm, and height 5 cm, failed at a compression load of 19.6 kN. Calculate the tensile strength (in MPa) of the rock specimen.	3																
3.	A cylindrical specimen of hard rock was subjected to diametral point load test. The pressure gauge reading at failure was 51 kg/cm². If the ram area was 14.426 cm², and the diameter & length of the core tested was 50 mm & 100mm respectively, calculate the uniaxial compressive strength and tensile strength of rock. (Ignore a size correction)	4																
4.	A rock sample is tested in uniaxial compression by connecting strain gages in axial and lateral directions. The results of the test are: uniaxial compressive strength (σ_c) = 42 MPa, Modulus of Elastisity (E) = 5.7 GPa and Poission's Ratio (ν) = 0.28. Determine the volumetric strain at the time of failure of the sample.	4																
5.	Data obtained from a series of triaxial test of shale rock is given below: <table><thead><tr><th>σ_3, MPa</th><th>σ_1, MPa</th></tr></thead><tbody><tr><td>0</td><td>25</td></tr><tr><td>0.5</td><td>28</td></tr><tr><td>1.0</td><td>33</td></tr><tr><td>1.5</td><td>37</td></tr><tr><td>2.0</td><td>40</td></tr><tr><td>3.0</td><td>49</td></tr><tr><td>4.0</td><td>57</td></tr></tbody></table>  Determine cohesion and angle of internal friction of the shale	σ_3 , MPa	σ_1 , MPa	0	25	0.5	28	1.0	33	1.5	37	2.0	40	3.0	49	4.0	57	5
σ_3 , MPa	σ_1 , MPa																	
0	25																	
0.5	28																	
1.0	33																	
1.5	37																	
2.0	40																	
3.0	49																	
4.0	57																	
6.	Calculate the shear strength (kN/m²) in terms of effective stress of soil on a plane within a saturated soil mass at a point. Assume the total normal stress is 290 kPa and pore water pressure is 105 kPa when cohesion (c') = 14 kPa and angle of internal friction is 30°.	4																
7.	A hydraulic fracture test in a granite rockmass yield the following results : <table><thead><tr><th>Depth (m)</th><th>Breakdown pressure, P_B (MPa)</th><th>Shut in pressure, P_S (MPa)</th></tr></thead><tbody><tr><td>500</td><td>14</td><td>8</td></tr></tbody></table> Assume the Tensile strength of the rock is 10 MPa and Unit weight of rock is 27 kN/m³. Estimate the values of σ_1, σ_2, and σ_3 assuming one principal stress is vertical and the pressure value is adjusted to account for the formation pressures (i.e $P_0 = 0$ for calculation purposes).	Depth (m)	Breakdown pressure, P_B (MPa)	Shut in pressure, P_S (MPa)	500	14	8	5										
Depth (m)	Breakdown pressure, P_B (MPa)	Shut in pressure, P_S (MPa)																
500	14	8																

END-SEMESTER EXAMINATION, Winter Semester, SESSION: 2022-23

Subject: Underground Metal Mining (MNC207), IV BTMNE

Answer any five Questions.

Maximum marks: 100

Time: 3:00hrs

Sl. No.	Description of Question	Marks
1.0	a) Briefly discuss the salient features of the operational underground metal mines in India.	10
	b) Characterize the primary and secondary development of underground metal mine based on the Inclination, Location, shape, size, and purposes.	10
2.0	a) Briefly Discuss on the following: i. Factors influencing development requirements ii. Different Purposes served by the mine development iii. Excavation Methods and its advantages & Disadvantages	10
	b) Enumerate the different methods of direction and gradient control of a drift. Briefly discuss the working principle of automated guidance system for direction and gradient control used in TBM.	10
3.0	A drift of 100 m length and opening size 4.5m x 3.2 m is to be driven from Point "A" to "B" in an underground metal mine using drilling and blasting method of excavation. Drift is to be developed using trackless mining system with single boom drill jumbo of 3.2 m drill rod, LHD and LPDT combination of 5 tonne and 15 tonne capacity, respectively, etc. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Steps involved in the development of the Drift ii. Manpower distribution plan activity wise/shift-wise iii. Days required to complete the development of the drift iv. Equipment/Machineries deployed/needed v. Drilling and blasting pattern vi. Drilling meterage per round of blasting vii. Explosive consumption per round of blasting viii. Tonnage produced per round of blasting ix. Powder factor obtained x. Ventilation circuit for ventilating the face	20
4.0	a) Why raising is preferred over winzing? Briefly discuss the different raising methods.	5
	b) An end raise/ventilation raise of 100m length and opening size 2m x 2m is to be developed in between two levels using Alimak Raise Climber (ARC). Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Steps involved in the raising process using ARC	15

DEPARTMENT OF MINING ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

Mid-Semester Examination

Examination: IV B. Tech. (Mining Engineering)

Semester: Winter

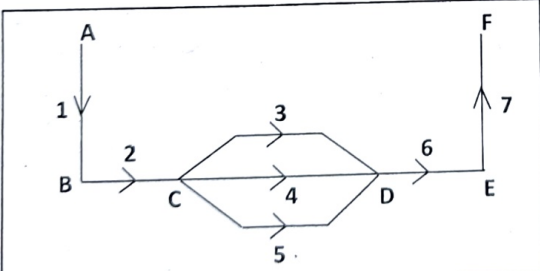
Session: 2021-22

Subject: Mine Ventilation (MNC 206)

Time: 1 Hour

Instruction: Answer ALL questions

Total Marks: 26

Q. No.	Question	Marks
1.	Discuss the different sources of heat in underground coal mines. Or With a neat sketch discuss the constructional features and working of a vapour-compression refrigeration system used in mine air-conditioning.	10
2.	In a mine ventilation circuit, there are two ventilation splits 'A' and 'B'. Split 'A' draws double the air quantity drawn by split 'B'. Resistance of split 'B' is exactly same as the resistance of the shafts and trunk airways. The quantity of air flowing in both the splits has to be made equal by putting a regulator in one of the splits. The main mine fan produces a pressure of 1000 Pa and supplies 100 m ³ /s of air in the mine. Calculate the size of the regulator. Assume the fan pressure remains same before and after installation of the regulator.	8
3.	The total quantity of air flowing through a mine as depicted below is 50 m³/s. The airways and node numbers have been indicated in the schematic diagram. The resistance (in N s²/m⁸) of the airways are given as follows: $R_1 = 0.027$, $R_2 = 0.397$, $R_3 = 3.578$, $R_4 = 2.091$, $R_5 = 5.590$, $R_6 = 0.199$, $R_7 = 0.684$. Determine the equivalent resistance of the mine airways and quantity of air flowing through each airway. 	8

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INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

Exam: IV B. Tech (Mining Engineering)

Session: 2022-23

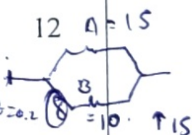
Semester: Winter

Subject: MINE VENTILATION (MNC 206)

Time: 3 hours

Total Marks: 100

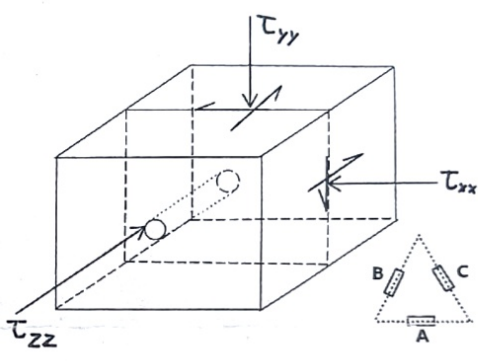
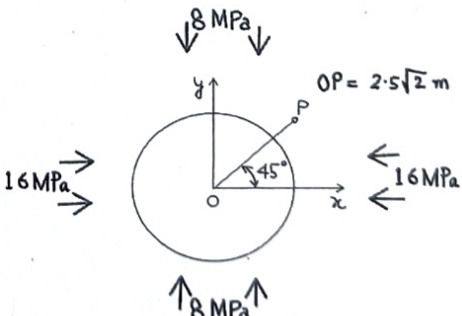
Instruction: Answer any FIVE questions

Q. 1	a) Discuss the generation sources of NO_x in underground mine atmosphere and its physiological effects on the human being. b) A 2.5 m thick gassy coal seam is occurring at a depth of 500 m from the surface. With neat sketches, explain the direct method of measurement of gas content of the coal seam.	3 + 3 14
Q. 2	a) Discuss the method of sampling of air in underground mine atmosphere and its analysis using a Gas Chromatograph. b) A 2000 m long and 200 m wide longwall panel is being worked by retreating method. With a neat sketch, discuss the methane drainage technique using surface boreholes for extraction of methane accumulated in the goaf of the longwall panel.	4 + 6 10
Q. 3	a) Discuss the applications of booster fan and auxiliary fan. b) A mine has three splits A, B and C, each of resistance 0.1 Ns²/m⁸. The trunk airways have a total resistance of 0.2 Ns²/m⁸. If the desired air quantity in splits A and B is 25 m³/s and 15 m³/s, respectively, calculate the airflow in split C when the surface fan generates a pressure of 700 Pa. Also, determine the specifications of the flow control devices in splits A and B to achieve these specified air quantities.	3 + 3 14 
Q. 4	a) With the help of a sketch, critically analyze how the change in direction of airflow due to natural ventilation influences the operation of main mine ventilation fan. b) A mine fan generates 500 Pa pressure and circulates 25 m³/s of air through the mine consisting of two splits, A and B. Split A and B circulates 15 m³/s and 10 m³/s of air, respectively. It is desired to increase the quantity of airflow in split B from 10 m³/s to 15 m³/s by installing a booster fan in it. Calculate the size of the booster fan if the resistance of the shafts and trunk airways is 0.2 Ns²/m⁸.	3 + 5 
Q. 5	a) With the help of a schematic mine ventilation network, discuss the purpose of the following ventilation control devices: (a) Regulator, and (b) Air-crossing. b) A horizontal underground mine gallery of 3.5 m wide, 2.5 m high and 700 m long lying at a depth of 250 m is naturally ventilated by means of two vertical shafts of 5.0 m diameter each. The gallery contains a normal bend with deflection of $\pi/3$ radians and radius of curvature of 3 m. The temperature measurements in the shafts are as follows: Downcast shaft-top = 293 K, Downcast shaft-bottom = 296 K Upcast shaft-top = 303 K, Upcast shaft-bottom = 304 K The coefficient of friction k for the shafts is 0.01 Ns²/m⁴ and for the underground gallery is 0.02 Ns²/m⁴. If the average barometric pressure in the shafts is 101.33 kPa, calculate the velocity of airflow in the gallery due to natural ventilation. Neglect the shock loss in the shafts.	6 14
Q. 6	a) With a neat sketch discuss the constructional features and working of an axial-flow fan, and mention its merits and limitations over the centrifugal fan. b) With a neat sketch, briefly discuss the mine characteristic curve, P-Q characteristic curve of fan, and operating point of fan.	8 + 4 2 + 6

Examination: MID SEMESTER
Subject: MNC 205 ROCK MECHANICS

Session: 2020-21
Semester: Winter
Time: 1 HOUR
Max. Marks: 30

- Instructions: 1. Answer ANY TWO questions
2. Neat and accurate sketches will assist you in solving the problems logically and faster
3. Assume logical data, where necessary

Q. No.	Question	Marks
1	(a) Use neat and labeled sketches to describe in brief the step-by step procedure adopted for determination of in situ stress using CSIR door-stopper cell. (b) Describe the schematic construction of a door-stopper cell using neat diagram. (c) What are the limitations of the method of in situ stress determination using CSIR door-stopper cell. (d) The following state of strain exists at a point 'P'. $[\epsilon_{ij}] = \begin{bmatrix} 0.03 & 0.04 & 0 \\ 0.04 & 0.06 & 0.05 \\ 0 & 0.05 & 0 \end{bmatrix}$ A line AB having direction cosines $n_x = 0.6$ $n_y = 0.8$ $n_z = 0$ is given. Determine the strain along the line AB and the volumetric strain at point P.	5+ 4+ 1+ 4+ 1 = (15)
2	A door-stopper cell consisting of an equilateral triangular strain rosette was utilised for in-situ stress measurement. The configuration of the strain gauges are shown in the figure below. The strain gauge 'A' was aligned along the x-direction while fixing at the end of the borehole wall. The maximum strains recorded by the strain gauges A, B and C are 900, 375 and 375 micro-strains respectively. The constants in the matrix relating the far field stress at the end of the borehole wall and other constants are given as under. $a = 1.25$ $b = -0.064$ $c = -0.75(0.645 + \nu)$ $d = 1.25$ $E = 6.5 \text{ GPa}$ and $\nu = 0.33$ Determine the 2-D state of stress. 	(15)
3	(a) Discuss the Coulomb failure criteria using a neat sketch of Mohr's circle. Then show that in terms of the principal stresses the criteria can also be expressed as $\sigma_1 = \frac{2c \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$ (b) A circular shaft of 5.0 m diameter is being excavated for mining. At a certain depth the shaft section is subjected to in situ stress as shown in the figure. The failure of the rock mass is governed by the failure criteria $\sigma_1 = \sigma_3 + \sqrt{m \sigma_c \sigma_3 + \sigma_c^2}$ where, $\sigma_c = 52 \text{ MPa}$ and $m = 7.5$. Use the Mohr circle concept to find the FoS at point P. 	7+ 8 = (15)
4	(a) Using neat sketches explain the procedure for determination of Liquid limit of soil using Cassagrande apparatus. (b) A soil contaminated with gasoline (of specific gravity = 0.9) with the following characteristics: $\rho_s = 2.65 \text{ Mg/m}^3$, $w = 25\%$, volume of the gasoline is 20% of the volume of the water, and 85% of the void space is filled with gasoline and water. - Complete the phase diagram and write their values - Find the void ratio and porosity of the specimen. - Find the total (Bulk) and dry density of the specimen.	5+ 6+ 2+ 2 = (15)

MID-SEMESTER EXAMINATION, Winter Semester, SESSION: 2022-23

Subject: Underground Metal Mining (MNC207), IV BTMNE

Answer All the Questions.

Maximum marks: 28

Time: 2:00hrs

Sl. No.	Description of Question	Marks
1.0	Briefly discuss about the operational underground metal mines in India	7
2.0	Characterize the primary and secondary development of underground metal mine based on the <u>Inclination, Location, shape, size, and purposes.</u>	7
3.0	Briefly Discuss on the following: a) Factors influencing development requirements b) Different Purposes served by the mine development c) Excavation Methods and its advantages & Disadvantages	7
4.0	A drift is to be driven from Point "A" to "B" in an underground metal mine. Briefly Explain, with neat sketch (wherever required), the steps involved in the development of the Drift using <u>drilling and Blasting method of excavation.</u>	7

Department of Mining Engineering
IIT(ISM), Dhanbad
Class Test

Semester: IV Sem (ME)

Session 2022-23

Subject: Underground Coal Mining (MNC 208)

Time 40 minutes

Marks : 20

Qn. No.	Question	Marks
1	Explain how two legged shield supports offer better stability at longwall face?	4
2	How do you calculate average support density / mean load density at maximum and minimum span of a longwall face? Assume your own data, if required.	5
3	Why debris on top of the canopies of longwall powered supports should be cleaned at regular intervals?	3
4	Using bulking factor theory, calculate the caving height of caving zone of a longwall face with 4 meter height of extraction.	4
5	Give two reasons why wider longwall faces are preferred.	4

DEPARTMENT OF MINING ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

Mid-Semester Examination

Examination: IV B. Tech. (Mining Engineering)

Semester: Winter

Session: 2022-23

Subject: Mine Ventilation (MNC 206)

Time: 2 Hours

Instruction: Answer ALL questions

Total Marks: 60

<i>Q. No.</i>	<i>Question</i>	<i>Marks</i>
1.	Discuss the different sources of heat in underground coal mines. OR With the help of a schematic diagram and superimposed pressure-enthalpy-state diagram, explain the constructional features and working of a vapour-compression refrigeration system used in mine air-conditioning.	15
2.	Explain the different modes of methane emission into underground coal mine atmosphere, and side by side discuss how to control such emission.	15
3.	Discuss about the generation of radon gas, its physiological effects and control measures in the mine environment.	10
4.	The analysis of air sample collected from an underground coal mine revealed the following: O ₂ = 16.52%, N ₂ = 77.93%, CO ₂ = 3.1%, CH ₄ = 2.44%, CO = 0.01%. Calculate the percentage of (i) Atmospheric air, (ii) Blackdamp, (iii) Firedamp, and (iv) Whitedamp in the mine air.	10
5.	The airflow rate in an ascensionally ventilated underground coal mine airway of 4.0 m width and 3.0 m height is 30 m ³ /s. If the rate of methane emission into the airway is 7.5 m ³ /s, calculate the methane layering number and critically analyse whether it is adequate to break the methane layering in the airway or not. What measures would you take to break the methane layering in this situation?	10

Time: 3 Hours

Max. Marks: 100

Instructions: 1) Answer any FIVE questions 2) Assume suitable data wherever necessary

Q. No	Question	Marks									
1.	At a depth of 450 m, a 3 m diameter circular tunnel is driven in rock having unit weight of 26 kN/m³ and uniaxial strength and tensile strengths of 40.0 MPa and 2.0 MPa, respectively. If the stress field ratio is 0.3. Find the induced stress at points (1), (2) and (3) as shown in the figure and also find the corresponding factor of safety. $\nu = 0.3$ (b) The stress in a granitic rock mass has been measured by the hydraulic fracturing technique. Two tests are conducted in a vertical borehole: one test at a depth of 800 m and the other test at a depth of 1200 m. The results are as follows: <table border="1"> <thead> <tr> <th>Depth (m)</th><th>Breakdown pressure, P_B (MPa)</th><th>Shut-in pressure, P_s (MPa)</th></tr> </thead> <tbody> <tr> <td>800</td><td>14.0</td><td>7.0</td></tr> <tr> <td>1200</td><td>24.5</td><td>19.0</td></tr> </tbody> </table> Given that the tensile strength of the rock is 10 MPa, estimate and list the values of σ_1, σ_2, and σ_3 at the two depths. Average unit weight of the rock mass is 26 kN/m³.	Depth (m)	Breakdown pressure, P_B (MPa)	Shut-in pressure, P_s (MPa)	800	14.0	7.0	1200	24.5	19.0	10
Depth (m)	Breakdown pressure, P_B (MPa)	Shut-in pressure, P_s (MPa)									
800	14.0	7.0									
1200	24.5	19.0									
2.	(a) Explain briefly the applications of Geomechanics classification of rock masses and Rock Mass Quality Index of rock mass classification systems in mining and rock engineering. (b) In a 1500-mm rock core run, the following rock pieces were recovered from a borehole: 53 mm, 108 mm, 125 mm, 75 mm, 148 mm, 320 mm, 68 mm, 145 mm, 35 mm and 134 mm. Find the RQD and the core recovery ratio. (c) A fine-grained soil is found to have a liquid limit of 90% and a plasticity index of 50. The natural water content is 28%. Determine the liquidity index and indicate the probable consistency of the natural soil.	5+5 5 5									
3.	(a) The RMR of immediate roof of a coal mine is estimated to be 42. The stress conditions at a given time in the roof are as follows: $\sigma_1 = 25 \text{ MPa}, \sigma_3 = 5 \text{ MPa}$ Considering m_i is 10 and uniaxial compressive strength (σ_{ci}) is 50 MPa. The coal mine galley is created by machine cutting i.e., completely avoiding blasting operations. Does this roof stratum fail? Explain your answer through calculations. (b) Discuss briefly the methods of reducing subsidence influence in mines.	10 10									

**Department of Mining Engineering
IIT(ISM), Dhanbad**

IV Semester B. Tech (Mining Engineering)

Mid-Semester Examination

Session 2022-23

Subject: Underground Coal Mining (MNC 208)

Time – 2 hours

Full marks - 60

All questions are compulsory

Question No.	Questions	Marks
1.	a) What do you understand by in-seam mining? In which conditions in-seam mining is preferred? Describe in brief, the advantages and disadvantages of horizon mining. <i>— generation process</i>	2+3+5
2.	Compare different underground coal mining methods like Bord & Pillar mining, Room & Pillar mining and Longwall mining with respect to the following factors: a) % age of extraction b) Effectiveness of strata control c) Risk of spontaneous heating <i>is more</i> d) Mass production capability <i>not from / in mind / more</i> e) Fleeting of face machinery	5x5
3.	Write short notes on the following: a) Breaker line support b) Hazards of shaft sinking <i>— slip</i> c) Diagonal line of extraction <i>step steep horizontal vertical</i>	5 x 3
4.	Determine the size of Bord and Pillar depillaring panel of a coal mine which can be extracted safely within the incubation period of 12 months with a rate of production of 500 Tonne per day. Thickness of the coal seam is 4.5 meter. Assume your own conditions as necessary.	10